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From: Steve Greer <[REDACTED]>
Sent: Sunday, March 25, 2018 8:04 PM
To: [REDACTED]
Subject: Fwd: Dr. Greer reply
Attachments: ancient-DNA-mutations.pdf

need these articles p and s

----- Forwarded message -----

From: red [REDACTED] >
Date: Mon, Mar 26, 2018 at 6:53 AM
Subject: RE: Dr. Greer reply
To: Steve Greer [REDACTED] >

Dear Dr. Greer (Steve), many thanks for your message and best wishes! I can talk anytime on Skype, but that medium is monitored just as closely as email (username "collie2" at this email address in Sydney, Australia).

Can you find articles on ancient DNA and its unreliability for extrapolation into operative genes?

I have attached **just one** of many literature papers about mutations due to age / oxidative damage in ancient DNA samples. These would seem sufficient to explain most/all of what the new authors report concerning the little mummy from Chile. The authors say "great care is required in interpretation" of the sequence data from ancient DNA for various reasons as described. I can find more papers if needed.

Are you a PhD or MD? And what has been your specialization ?

I am a Caltech Ph.D. and did the first x-ray crystal structures of DNA on Earth, and know also evolution, PCR, etc. but have not worked with ancient DNA samples personally. My last project was as part of a team that developed a new DNA test for colorectal cancer. I did full-genome studies to find out which human genes were modified in a high percentage of cancer tissue

samples versus normal tissue samples. Given that your emails may be monitored, I do not wish to say more here.

<http://www.pnas.org/content/78/4/2179> or <https://www.ncbi.nlm.nih.gov/pubmed/7402336> or <https://www.sciencedirect.com/science/book/9780121550899>

also <https://www.amazon.com/Hair-Alien-Forensic-Evidence-Abductions/dp/0743492862> (Bill Chalker lives nearby) or <https://www.youtube.com/watch?v=PxA8lQ5A3qQ> or <https://www.youtube.com/watch?v=7aAMhHn9Y9Y> or <https://www.youtube.com/watch?v=gA1gF8FtyoI>

At any rate, please let me know when we can speak. Generally 11pm-midnight eastern daylight time is best.

I will be glad to speak, but Skype is closely watched like email, if you are worried about that? It would be 14 hours later here in Sydney about 2 PM.

>>>My feeling , in order to respond effectively to this new paper, would be to say that “all ancient samples >100 years old have lots of oxidative damage caused by environmental exposure after death”, as noted in 20 different ancient DNA samples studied in the example paper attached. We know that DNA from the small Chilean mummy was moderately damaged in the same way, by its small average size (300 bp), and by finding a twofold excess of C>T mutations near the ends of those fragments (which generally mutate more easily than the centres, because loose ends of a double helix may fray into single strands). How then do we know that standard patterns of age-related mutation **after death**, in that small Chilean mummy which is 100-500 years old, were not responsible for the **vast number** of (sometimes novel) mutations which were found, rather than putative mutations which took place in a small fetus while it was **still alive**? Due hypothetically to “sodium nitrate exposure”, which has never been shown to lead to serious mutations except in laboratory model systems, and is a safe food additive eaten commonly by pregnant women today, in bacon or other meats?

In other words, we have one very-well documented kind of mutation (after death in ancient samples) which could explain most/all of their results nicely, whereas they postulate an entirely new mechanism (nitrate exposure during life) which has never been seen before except in some laboratory models, and has no other factual support whatsoever. Which interpretation seems the most plausible? *Can they prove in any way that the vast numbers of mutations which they see, are different from those found in all other ancient DNA samples? The two spectra of mutation are almost identical, it might be noted.*

And not say anything about "conspiracies". Most scientists should agree with the comments given above. You can test such arguments on various scientific friends there, to see what they think? I can find more literature papers if needed, but the one attached is a good example. At the very least, this will force the authors on the defensive, to try and defend their specious and unprecedented arguments, and you can see where the debate leads after that? Without having access to their huge original database, we cannot know what is really in it, or how they will respond? All we can go on is what they wrote in a peer-reviewed paper. News articles don't count....I would not have accepted this paper as a referee, unless they could answer adequately the serious technical and interpretative issues summarized above. It is never easy to respond to a faulty paper which has already been published. Most people will just assume that it was reviewed and approved for accuracy by the journal.

In one quick sentence, the authors need to provide some proof that the many novel mutations which say were responsible for "bone deformities" while the mummy was alive, did not in fact take place in the 100-500 years or so, since the mummy died? Since we see similar mutations in almost all samples of ancient DNA? I am pretty sure that is correct, since a little fetus could not develop to such a late, well-developed stage with 54 separate deleterious mutations in its developmental genes. Some expert at embryology might be able to tell you how early in pregnancy such a highly mutated embryo would abort.

Past studies of ancient DNA have generally relied on mitochondrial DNA, where these problems are not as vexing. The current authors are using ancient DNA to study nuclear genes, for which the problem of interpretation seem very serious indeed. Are the base changes which you see real or artefactual?

Going out for a while, 10 AM here and sunny! Best wishes,  in Sydney